

ROOTS-AI™ C-02 Canonical Scoring Rules & Golden Tests v1.0.1 - CORRECTED

Controlled human-readable companion to the authoritative executable XLSX. Deterministic scoring only; LLM has no calculation authority.

v1.0.1 closes driver-ranking/tie inconsistencies across the 30 Golden Tests, reserves BS exclusively for Biological Safety Signals™, defines BIO_STATE for Biological State™, unifies integer rounding as half-away-from-zero, and treats classification hex values as semantic references rather than final UI tokens.

README

Field	Value	Control	Implementation note	Owner	Status
Dataset ID	ROOTS-C02-SCORING-001	Immutable	Link every score/report to v1.0.1	ROOTS	Approved
scoring_version	1.0.1	Immutable	Server-side configuration only	ROOTS	Approved
Scored questions	40	Exact	All other C-01 questions are contextual-only	ROOTS	Approved
Raw burden scale	0-4	Exact	0=lowest burden; 4=highest burden	ROOTS	Approved
Domain coverage threshold	50%	Exact	Below threshold returns null	ROOTS	Approved
Biological State minimum	5 of 7 domains	Exact	Otherwise null	ROOTS	Approved
Rounding	Half away from zero to integer	Exact	Opportunity/Recovery retain one decimal	ROOTS	Approved
LLM role	None in calculation	Mandatory	LLM cannot change any score, class or driver	ROOTS	Approved

Domains

domain_id	display_name	meaning	formula
MR	Metabolic Resistance™	Self-reported weight-change resistance and activity context	ROUND(SUM(answer_points)/SUM(max_points)*100)
HS	Hunger & Satiety Signals™	Hunger, cravings, fullness and meal response	ROUND(SUM(answer_points)/SUM(max_points)*100)
SR	Sleep Recovery Index™	Sleep duration, continuity and restoration	ROUND(SUM(answer_points)/SUM(max_points)*100)
CH	Circadian Health Score™	Timing of light, screens, meals and sleep	ROUND(SUM(answer_points)/SUM(max_points)*100)
SL	Stress Load™	Perceived tension, stress-eating and cognitive activation	ROUND(SUM(answer_points)/SUM(max_points)*100)
IB	Inflammation Burden Index™	Non-specific self-reported symptom burden; not a biomarker	ROUND(SUM(answer_points)/SUM(max_points)*100)
BS	Biological Safety Signals™	Perceived resistance, appetite drive and low energy	ROUND(SUM(answer_points)/SUM(max_points)*100)

Question_Mapping

question_id	domain_id	weight	reverse_scored	points_map	question_reference	normalized_range	status
Q9	MR	1	False	GAIN_PATTERN	How did the weight change that concerns you begin?	0-4 burden points	APPROVED
Q10	MR	1	False	FREQ	How often have well-planned diet or activity efforts produced less change than you expected?	0-4 burden points	APPROVED
Q11	MR	1	False	FREQ	After an intentional weight loss, how often has some or all of the weight returned?	0-4 burden points	APPROVED
Q12	MR	1	False	WEIGHT_CYCLE S	How many meaningful cycles of weight loss and regain have you experienced?	0-4 burden points	APPROVED
Q46	MR	1	False	ACTIVITY_DAYS	On how many days per week do you complete intentional physical activity?	0-4 burden points	APPROVED
Q47	MR	1	False	ACTIVITY_DURATION	On active days, what is the usual duration of your intentional activity?	0-4 burden points	APPROVED
Q48	MR	1	False	ACTIVITY_TYPE	Which option best describes your usual physical activity pattern?	0-4 burden points	APPROVED
Q16	SR	1	False	SLEEP_HOURS	How many hours of sleep do you typically get in a 24-hour period?	0-4 burden points	APPROVED
Q17	SR	1	False	FREQ	How often do you feel tired even after what seemed like enough time in bed?	0-4 burden points	APPROVED
Q18	SR	1	False	FREQ	How often do you wake during the night and struggle to return to sleep?	0-4 burden points	APPROVED
Q19	SR	1	False	FREQ	How often is your bedtime after midnight?	0-4 burden points	APPROVED
Q20	SR	1	False	FREQ	How often do you remain awake until 2:00 AM or later because you cannot settle to sleep?	0-4 burden points	APPROVED
Q21	SR	1	False	FREQ	How often do you snore loudly, wake gasping, or receive feedback that your breathing pauses during sleep?	0-4 burden points	APPROVED
Q22	SR	1	False	FREQ	How often do you experience a marked afternoon energy crash?	0-4 burden points	APPROVED
Q23	HS	1	False	FREQ	How often do you feel physically hungry within two hours after a full meal?	0-4 burden points	APPROVED
Q24	HS	1	False	FREQ	How often do you experience food cravings when you are not physically hungry?	0-4 burden points	APPROVED
Q25	HS	1	False	FREQ	How often do you eat enough but still fail to feel comfortably full?	0-4 burden points	APPROVED
Q26	HS	1	True	FREQ	How often do you feel comfortably full after a normal meal?	0-4 burden points	APPROVED
Q27	HS	1	False	FREQ	How often are you hungry again three to four hours after a protein-rich meal?	0-4 burden points	APPROVED

question_id	domain_id	weight	reverse_scored	points_map	question_reference	normalized_range	status
Q28	HS	1	True	FREQ	How often does a protein-rich meal keep you satisfied for four hours or longer?	0-4 burden points	APPROVED
Q29	HS	1	False	FREQ	How often do you feel sleepy, foggy or low in energy after a main meal?	0-4 burden points	APPROVED
Q30	HS	1	False	FREQ	When you start eating sweets or snack foods, how often is it difficult to stop at the amount you intended?	0-4 burden points	APPROVED
Q31	SL	1	False	FREQ	How often do you feel persistent physical tension or stress?	0-4 burden points	APPROVED
Q32	SL	1	False	FREQ	How often does stress increase your desire to eat?	0-4 burden points	APPROVED
Q33	SL	1	False	FREQ	How often does your mind race when you are trying to sleep?	0-4 burden points	APPROVED
Q34	SL	1	False	FREQ	How often are cravings noticeably stronger in the evening?	0-4 burden points	APPROVED
Q35	IB	1	False	FREQ	How often do you experience bloating, gas or digestive discomfort?	0-4 burden points	APPROVED
Q36	IB	1	False	FREQ	How often do you experience brain fog after meals?	0-4 burden points	APPROVED
Q37	IB	1	False	FREQ	How often do you experience unexplained joint pain or stiffness?	0-4 burden points	APPROVED
Q38	IB	1	False	FREQ	How often does fat around your waist seem resistant to your usual efforts?	0-4 burden points	APPROVED
Q39	IB	1	False	FREQ	How often do you experience unusual thirst or a dry mouth?	0-4 burden points	APPROVED
Q40	IB	1	False	FREQ	How often have reducing calories and increasing activity produced little sustained change?	0-4 burden points	APPROVED
Q41	CH	1	False	FREQ	How often do you spend the first hour after waking indoors without natural daylight exposure?	0-4 burden points	APPROVED
Q42	CH	1	False	FREQ	How often do you use a phone, tablet or computer during the final hour before sleep?	0-4 burden points	APPROVED
Q43	CH	1	False	FREQ	How often do you eat within three hours of bedtime?	0-4 burden points	APPROVED
Q44	CH	1	False	FREQ	How often does your sleep schedule vary by more than two hours across the week?	0-4 burden points	APPROVED
Q45	CH	1	False	FREQ	How often do you feel more alert late at night than during the morning?	0-4 burden points	APPROVED
Q49	BS	1	False	FREQ	How often does your body seem to resist weight loss despite consistent effort?	0-4 burden points	APPROVED
Q50	BS	1	False	FREQ	How often does hunger feel stronger than your ability to regulate it?	0-4 burden points	APPROVED

question_id	domain_id	weight	reverse_scored	points_map	question_reference	normalized_range	status
Q51	BS	1	False	FREQ	How often are your energy levels lower than you believe they should be?	0-4 burden points	APPROVED

Option_Points

question_id	domain_id	option_set_id	option_id	display_label	burden_points	excluded_as_na
Q9	MR	GAIN_PATTERN	STABLE	No concerning change or mostly stable	0	False
Q9	MR	GAIN_PATTERN	IDENTIFIABLE	After an identifiable life period or event	1	False
Q9	MR	GAIN_PATTERN	GRADUAL	Gradually with no single clear reason	2	False
Q9	MR	GAIN_PATTERN	CYCLING	In repeated cycles of loss and regain	3	False
Q9	MR	GAIN_PATTERN	RAPID	Rapidly or unexpectedly	4	False
Q9	MR	GAIN_PATTERN	NA	Not applicable		True
Q10	MR	FREQ	NVR	Never	0	False
Q10	MR	FREQ	RLY	Rarely	1	False
Q10	MR	FREQ	SMT	Sometimes	2	False
Q10	MR	FREQ	OFT	Often	3	False
Q10	MR	FREQ	ALW	Almost always	4	False
Q10	MR	FREQ	NA	Not applicable		True
Q11	MR	FREQ	NVR	Never	0	False
Q11	MR	FREQ	RLY	Rarely	1	False
Q11	MR	FREQ	SMT	Sometimes	2	False
Q11	MR	FREQ	OFT	Often	3	False
Q11	MR	FREQ	ALW	Almost always	4	False
Q11	MR	FREQ	NA	Not applicable		True
Q12	MR	WEIGHT_CYCLES	NONE	None	0	False
Q12	MR	WEIGHT_CYCLES	ONE	One	1	False
Q12	MR	WEIGHT_CYCLES	TWO	Two	2	False
Q12	MR	WEIGHT_CYCLES	THREE_FOUR	Three or four	3	False
Q12	MR	WEIGHT_CYCLES	FIVE_PLUS	Five or more	4	False
Q12	MR	WEIGHT_CYCLES	NA	Not applicable		True
Q46	MR	ACTIVITY_DAYS	D0	0 days	4	False
Q46	MR	ACTIVITY_DAYS	D1_2	1-2 days	3	False
Q46	MR	ACTIVITY_DAYS	D3_4	3-4 days	1	False
Q46	MR	ACTIVITY_DAYS	D5_7	5-7 days	0	False
Q46	MR	ACTIVITY_DAYS	NA	Not applicable		True
Q47	MR	ACTIVITY_DURATION	LT10	Less than 10 minutes	4	False
Q47	MR	ACTIVITY_DURATION	M10_29	10-29 minutes	3	False
Q47	MR	ACTIVITY_DURATION	M30_59	30-59 minutes	1	False
Q47	MR	ACTIVITY_DURATION	M60_PLUS	60 minutes or more	0	False
Q47	MR	ACTIVITY_DURATION	NA	Not applicable		True
Q48	MR	ACTIVITY_TYPE	NONE	No intentional activity	4	False
Q48	MR	ACTIVITY_TYPE	LIGHT	Light movement or daily activities	3	False
Q48	MR	ACTIVITY_TYPE	AEROBIC	Mostly aerobic activity	2	False
Q48	MR	ACTIVITY_TYPE	RESISTANCE	Mostly resistance training	1	False
Q48	MR	ACTIVITY_TYPE	MIXED	A mix of aerobic and resistance activity	0	False

question_id	domain_id	option_set_id	option_id	display_label	burden_points	excluded_as_na
Q48	MR	ACTIVITY_TYPE	NA	Not applicable		True
Q16	SR	SLEEP_HOURS	LT5	Less than 5 hours	4	False
Q16	SR	SLEEP_HOURS	H5_6	5 to less than 6 hours	3	False
Q16	SR	SLEEP_HOURS	H6_7	6 to less than 7 hours	1	False
Q16	SR	SLEEP_HOURS	H7_9	7 to 9 hours	0	False
Q16	SR	SLEEP_HOURS	GT9	More than 9 hours	2	False
Q16	SR	SLEEP_HOURS	NA	Not applicable		True
Q17	SR	FREQ	NVR	Never	0	False
Q17	SR	FREQ	RLY	Rarely	1	False
Q17	SR	FREQ	SMT	Sometimes	2	False
Q17	SR	FREQ	OFT	Often	3	False
Q17	SR	FREQ	ALW	Almost always	4	False
Q17	SR	FREQ	NA	Not applicable		True
Q18	SR	FREQ	NVR	Never	0	False
Q18	SR	FREQ	RLY	Rarely	1	False
Q18	SR	FREQ	SMT	Sometimes	2	False
Q18	SR	FREQ	OFT	Often	3	False
Q18	SR	FREQ	ALW	Almost always	4	False
Q18	SR	FREQ	NA	Not applicable		True
Q19	SR	FREQ	NVR	Never	0	False
Q19	SR	FREQ	RLY	Rarely	1	False
Q19	SR	FREQ	SMT	Sometimes	2	False
Q19	SR	FREQ	OFT	Often	3	False
Q19	SR	FREQ	ALW	Almost always	4	False
Q19	SR	FREQ	NA	Not applicable		True
Q20	SR	FREQ	NVR	Never	0	False
Q20	SR	FREQ	RLY	Rarely	1	False
Q20	SR	FREQ	SMT	Sometimes	2	False
Q20	SR	FREQ	OFT	Often	3	False
Q20	SR	FREQ	ALW	Almost always	4	False
Q20	SR	FREQ	NA	Not applicable		True
Q21	SR	FREQ	NVR	Never	0	False
Q21	SR	FREQ	RLY	Rarely	1	False
Q21	SR	FREQ	SMT	Sometimes	2	False
Q21	SR	FREQ	OFT	Often	3	False
Q21	SR	FREQ	ALW	Almost always	4	False
Q21	SR	FREQ	NA	Not applicable		True
Q22	SR	FREQ	NVR	Never	0	False
Q22	SR	FREQ	RLY	Rarely	1	False
Q22	SR	FREQ	SMT	Sometimes	2	False
Q22	SR	FREQ	OFT	Often	3	False
Q22	SR	FREQ	ALW	Almost always	4	False
Q22	SR	FREQ	NA	Not applicable		True
Q23	HS	FREQ	NVR	Never	0	False
Q23	HS	FREQ	RLY	Rarely	1	False
Q23	HS	FREQ	SMT	Sometimes	2	False
Q23	HS	FREQ	OFT	Often	3	False
Q23	HS	FREQ	ALW	Almost always	4	False

question_id	domain_id	option_set_id	option_id	display_label	burden_points	excluded_as_na
Q23	HS	FREQ	NA	Not applicable		True
Q24	HS	FREQ	NVR	Never	0	False
Q24	HS	FREQ	RLY	Rarely	1	False
Q24	HS	FREQ	SMT	Sometimes	2	False
Q24	HS	FREQ	OFT	Often	3	False
Q24	HS	FREQ	ALW	Almost always	4	False
Q24	HS	FREQ	NA	Not applicable		True
Q25	HS	FREQ	NVR	Never	0	False
Q25	HS	FREQ	RLY	Rarely	1	False
Q25	HS	FREQ	SMT	Sometimes	2	False
Q25	HS	FREQ	OFT	Often	3	False
Q25	HS	FREQ	ALW	Almost always	4	False
Q25	HS	FREQ	NA	Not applicable		True
Q26	HS	FREQ	NVR	Never	4	False
Q26	HS	FREQ	RLY	Rarely	3	False
Q26	HS	FREQ	SMT	Sometimes	2	False
Q26	HS	FREQ	OFT	Often	1	False
Q26	HS	FREQ	ALW	Almost always	0	False
Q26	HS	FREQ	NA	Not applicable		True
Q27	HS	FREQ	NVR	Never	0	False
Q27	HS	FREQ	RLY	Rarely	1	False
Q27	HS	FREQ	SMT	Sometimes	2	False
Q27	HS	FREQ	OFT	Often	3	False
Q27	HS	FREQ	ALW	Almost always	4	False
Q27	HS	FREQ	NA	Not applicable		True
Q28	HS	FREQ	NVR	Never	4	False
Q28	HS	FREQ	RLY	Rarely	3	False
Q28	HS	FREQ	SMT	Sometimes	2	False
Q28	HS	FREQ	OFT	Often	1	False
Q28	HS	FREQ	ALW	Almost always	0	False
Q28	HS	FREQ	NA	Not applicable		True
Q29	HS	FREQ	NVR	Never	0	False
Q29	HS	FREQ	RLY	Rarely	1	False
Q29	HS	FREQ	SMT	Sometimes	2	False
Q29	HS	FREQ	OFT	Often	3	False
Q29	HS	FREQ	ALW	Almost always	4	False
Q29	HS	FREQ	NA	Not applicable		True
Q30	HS	FREQ	NVR	Never	0	False
Q30	HS	FREQ	RLY	Rarely	1	False
Q30	HS	FREQ	SMT	Sometimes	2	False
Q30	HS	FREQ	OFT	Often	3	False
Q30	HS	FREQ	ALW	Almost always	4	False
Q30	HS	FREQ	NA	Not applicable		True
Q31	SL	FREQ	NVR	Never	0	False
Q31	SL	FREQ	RLY	Rarely	1	False
Q31	SL	FREQ	SMT	Sometimes	2	False
Q31	SL	FREQ	OFT	Often	3	False
Q31	SL	FREQ	ALW	Almost always	4	False

question_id	domain_id	option_set_id	option_id	display_label	burden_points	excluded_as_na
Q31	SL	FREQ	NA	Not applicable		True
Q32	SL	FREQ	NVR	Never	0	False
Q32	SL	FREQ	RLY	Rarely	1	False
Q32	SL	FREQ	SMT	Sometimes	2	False
Q32	SL	FREQ	OFT	Often	3	False
Q32	SL	FREQ	ALW	Almost always	4	False
Q32	SL	FREQ	NA	Not applicable		True
Q33	SL	FREQ	NVR	Never	0	False
Q33	SL	FREQ	RLY	Rarely	1	False
Q33	SL	FREQ	SMT	Sometimes	2	False
Q33	SL	FREQ	OFT	Often	3	False
Q33	SL	FREQ	ALW	Almost always	4	False
Q33	SL	FREQ	NA	Not applicable		True
Q34	SL	FREQ	NVR	Never	0	False
Q34	SL	FREQ	RLY	Rarely	1	False
Q34	SL	FREQ	SMT	Sometimes	2	False
Q34	SL	FREQ	OFT	Often	3	False
Q34	SL	FREQ	ALW	Almost always	4	False
Q34	SL	FREQ	NA	Not applicable		True
Q35	IB	FREQ	NVR	Never	0	False
Q35	IB	FREQ	RLY	Rarely	1	False
Q35	IB	FREQ	SMT	Sometimes	2	False
Q35	IB	FREQ	OFT	Often	3	False
Q35	IB	FREQ	ALW	Almost always	4	False
Q35	IB	FREQ	NA	Not applicable		True
Q36	IB	FREQ	NVR	Never	0	False
Q36	IB	FREQ	RLY	Rarely	1	False
Q36	IB	FREQ	SMT	Sometimes	2	False
Q36	IB	FREQ	OFT	Often	3	False
Q36	IB	FREQ	ALW	Almost always	4	False
Q36	IB	FREQ	NA	Not applicable		True
Q37	IB	FREQ	NVR	Never	0	False
Q37	IB	FREQ	RLY	Rarely	1	False
Q37	IB	FREQ	SMT	Sometimes	2	False
Q37	IB	FREQ	OFT	Often	3	False
Q37	IB	FREQ	ALW	Almost always	4	False
Q37	IB	FREQ	NA	Not applicable		True
Q38	IB	FREQ	NVR	Never	0	False
Q38	IB	FREQ	RLY	Rarely	1	False
Q38	IB	FREQ	SMT	Sometimes	2	False
Q38	IB	FREQ	OFT	Often	3	False
Q38	IB	FREQ	ALW	Almost always	4	False
Q38	IB	FREQ	NA	Not applicable		True
Q39	IB	FREQ	NVR	Never	0	False
Q39	IB	FREQ	RLY	Rarely	1	False
Q39	IB	FREQ	SMT	Sometimes	2	False
Q39	IB	FREQ	OFT	Often	3	False
Q39	IB	FREQ	ALW	Almost always	4	False

question_id	domain_id	option_set_id	option_id	display_label	burden_points	excluded_as_na
Q39	IB	FREQ	NA	Not applicable		True
Q40	IB	FREQ	NVR	Never	0	False
Q40	IB	FREQ	RLY	Rarely	1	False
Q40	IB	FREQ	SMT	Sometimes	2	False
Q40	IB	FREQ	OFT	Often	3	False
Q40	IB	FREQ	ALW	Almost always	4	False
Q40	IB	FREQ	NA	Not applicable		True
Q41	CH	FREQ	NVR	Never	0	False
Q41	CH	FREQ	RLY	Rarely	1	False
Q41	CH	FREQ	SMT	Sometimes	2	False
Q41	CH	FREQ	OFT	Often	3	False
Q41	CH	FREQ	ALW	Almost always	4	False
Q41	CH	FREQ	NA	Not applicable		True
Q42	CH	FREQ	NVR	Never	0	False
Q42	CH	FREQ	RLY	Rarely	1	False
Q42	CH	FREQ	SMT	Sometimes	2	False
Q42	CH	FREQ	OFT	Often	3	False
Q42	CH	FREQ	ALW	Almost always	4	False
Q42	CH	FREQ	NA	Not applicable		True
Q43	CH	FREQ	NVR	Never	0	False
Q43	CH	FREQ	RLY	Rarely	1	False
Q43	CH	FREQ	SMT	Sometimes	2	False
Q43	CH	FREQ	OFT	Often	3	False
Q43	CH	FREQ	ALW	Almost always	4	False
Q43	CH	FREQ	NA	Not applicable		True
Q44	CH	FREQ	NVR	Never	0	False
Q44	CH	FREQ	RLY	Rarely	1	False
Q44	CH	FREQ	SMT	Sometimes	2	False
Q44	CH	FREQ	OFT	Often	3	False
Q44	CH	FREQ	ALW	Almost always	4	False
Q44	CH	FREQ	NA	Not applicable		True
Q45	CH	FREQ	NVR	Never	0	False
Q45	CH	FREQ	RLY	Rarely	1	False
Q45	CH	FREQ	SMT	Sometimes	2	False
Q45	CH	FREQ	OFT	Often	3	False
Q45	CH	FREQ	ALW	Almost always	4	False
Q45	CH	FREQ	NA	Not applicable		True
Q49	BS	FREQ	NVR	Never	0	False
Q49	BS	FREQ	RLY	Rarely	1	False
Q49	BS	FREQ	SMT	Sometimes	2	False
Q49	BS	FREQ	OFT	Often	3	False
Q49	BS	FREQ	ALW	Almost always	4	False
Q49	BS	FREQ	NA	Not applicable		True
Q50	BS	FREQ	NVR	Never	0	False
Q50	BS	FREQ	RLY	Rarely	1	False
Q50	BS	FREQ	SMT	Sometimes	2	False
Q50	BS	FREQ	OFT	Often	3	False
Q50	BS	FREQ	ALW	Almost always	4	False

question_id	domain_id	option_set_id	option_id	display_label	burden_points	excluded_as_na
Q50	BS	FREQ	NA	Not applicable		True
Q51	BS	FREQ	NVR	Never	0	False
Q51	BS	FREQ	RLY	Rarely	1	False
Q51	BS	FREQ	SMT	Sometimes	2	False
Q51	BS	FREQ	OFT	Often	3	False
Q51	BS	FREQ	ALW	Almost always	4	False
Q51	BS	FREQ	NA	Not applicable		True

Formulas

rule_id	output	normative_formula	null_or_boundary_rule	rounding	audit_fields
SC-001	Domain score	$100 \times \Sigma(\text{points} \times \text{weight}) \div \Sigma(4 \times \text{weight})$, answered eligible items only	N/A/missing excluded; null if answered/eligible < 0.50	Half away from zero to integer	question IDs, raw option IDs, points, weights, numerator, denominator
SC-002	Biological State™	Mean of available seven domain scores	Null if fewer than 5 domains available	Half away from zero to integer	available/null domains and mean
SC-003	Opportunity Score™	$100 - (\text{Biological State} \times 0.5)$	Null when Biological State is null; clamp 0-100	One decimal	Biological State and constant 0.5
SC-004	Protective Factor Score	$20 \times \text{count of active P1-P5}$	Count 0-5; no imputation	Half away from zero to integer	factor IDs and booleans
SC-005	Recovery Potential™	$0.30 \times (100 - \text{BIO_STATE}) + 0.25 \times \text{Protective} + 0.20 \times \text{Age} + 0.15 \times \text{Condition} + 0.10 \times \text{Medication}$	Null when Biological State is null; clamp 0-100	One decimal	BIO_STATE, all factor values and contributions
SC-006	Coverage	$\text{Answered scored items} \div \text{eligible scored items} \times 100$	N/A and missing count as not answered for confidence coverage	Half away from zero to integer	answered/eligible counts
SC-007	Consistency by domain	$\max(0, 1 - \text{populationSD}(\text{points})/2) \times 100$	Null when domain is null	Half away from zero to integer	points, mean, SD
SC-008	ROOTS Confidence™	$0.50 \times \text{overall coverage} + 0.30 \times \text{Q72 value} + 0.20 \times \text{mean available-domain consistency}$	0 when no consistency; never changes scores	Half away from zero to integer	three components

Protective_Factors

factor_id	source	activation/value rule	output_value	purpose	notes
P1	Q46/Q47	Activity ≥3 days/week AND usual duration ≥30 minutes	20	Protective score	Use option IDs D3_4/D5_7 and M30_59/M60_PLUS
P2	Q64	Often or Almost always	20	Protective score	Meal timing consistency
P3	Q65	Supportive or Very supportive	20	Protective score	Home environment
P4	Q61	Never or Former use	20	Protective score	Nicotine context
P5	Q69	Readiness 7-10	20	Protective score	Readiness
AGE	Q1	16-30=100; 31-45=80; 46-60=60; 61-75=40; 76-110=20	Band value	Recovery Potential	16-17 uses 18-30 band if participation is enabled
CONDITION	Q13	None=100; one category=75; two=50; three or more=25	Band value	Recovery Potential	PREFER_NOT/NA treated as unavailable and sets Recovery Potential limitation flag
MEDICATION	Q14	None=100; one category=75; two or more=50	Band value	Recovery Potential	UNSURE/PREFER_NOT/NA sets limitation flag; no medication advice

Classifications

scale	minimum	maximum	label	color_hex	approved_interpretation
DOMAIN	0	24	Optimized	#27AE60	Lower reported burden
DOMAIN	25	49	Compensating	#F39C12	Signals present; compensation appears active
DOMAIN	50	74	Strained	#E67E22	Multiple or frequent signals
DOMAIN	75	100	Dysregulated	#C0392B	High reported burden; educational prompt to seek appropriate support
CONFIDENCE	80	100	High	#27AE60	Strong data completeness and consistency
CONFIDENCE	60	79	Moderate-High	#F39C12	Good data with some limitations
CONFIDENCE	40	59	Moderate	#E67E22	Interpret cautiously
CONFIDENCE	0	39	Low	#C0392B	Substantial limitations
RECOVERY	75	100	High	#27AE60	Many modifiable/protective features
RECOVERY	50	74	Moderate	#F39C12	Meaningful recovery opportunities
RECOVERY	25	49	Limited	#E67E22	Progress may require more support
RECOVERY	0	24	Low	#C0392B	Educational prompt for professional support

Drivers_Evidence

rule_id	subject	deterministic rule	output	tie_or_fallback	audit requirement
DRV-001	Driver eligibility	Available domain score ≥25	Eligible	Scores <25 are not drivers	Record all candidates
DRV-002	Ranking	Sort eligible domains by score descending; exact ties use fixed order MR,HS,SR,CH,SL,IB,BS	Primary/secondary/tertiary	No LLM selection; null/<25 domains excluded	Record ordered list
DRV-003	Co-primary	If top two eligible scores differ by ≤3, report them once as one co-primary pair	Co-primary pair, then next highest-ranked distinct eligible domain if one exists	No duplication; no ineligible fallback; preserve canonical tie order	Record tie delta
DRV-004	No dominant burden	No domain ≥25	No primary driver	Display strengths-focused copy	Record reason
EVD-001	Evidence score	0.45×domain coverage + 0.35×domain consistency + 0.20×high-signal proportion	0-100	High-signal means points ≥3	Trace components
EVD-002	Evidence label	80-100 Strong; 60-79 Moderate; 40-59 Limited; 0-39 Weak	Label	Not clinical evidence grade	Record boundary
CNF-001	Confidence	Formula SC-008	High/Moderate-High/Moderate/Low	Never modifies scores	Trace components

Golden_Tests

test_id	purpose	normalized_input_json	expected_output_json
GT-001	All scored items at minimum burden	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":0,"SR":0,"HS":0,"SL":0,"IB":0,"CH":0,"BS":0},"coverage":{ "MR...
GT-002	All scored items at maximum burden	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":100,"SR":100,"HS":100,"SL":100,"IB":100,"CH":100,"BS":100},"...
GT-003	All scored items at midpoint	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":50,"SR":50,"HS":50,"SL":50,"IB":50,"CH":50,"BS":50},"coverag...
GT-004	MR only at maximum	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":100,"SR":0,"HS":0,"SL":0,"IB":0,"CH":0,"BS":0},"coverage":{ "...
GT-005	SR only at maximum	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":0,"SR":100,"HS":0,"SL":0,"IB":0,"CH":0,"BS":0},"coverage":{ "...
GT-006	HS only at maximum	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":0,"SR":0,"HS":100,"SL":0,"IB":0,"CH":0,"BS":0},"coverage":{ "...
GT-007	SL only at maximum	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":0,"SR":0,"HS":0,"SL":100,"IB":0,"CH":0,"BS":0},"coverage":{ "...
GT-008	IB only at maximum	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":0,"SR":0,"HS":0,"SL":0,"IB":100,"CH":0,"BS":0},"coverage":{ "...
GT-009	CH only at maximum	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":0,"SR":0,"HS":0,"SL":0,"IB":0,"CH":100,"BS":0},"coverage":{ "...
GT-010	BS only at maximum	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":0,"SR":0,"HS":0,"SL":0,"IB":0,"CH":0,"BS":100},"coverage":{ "...
GT-011	Domain classification boundary 25	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":50,"SR":50,"HS":50,"SL":25,"IB":50,"CH":50,"BS":50},"coverag...
GT-012	Domain classification boundary 50	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":50,"SR":50,"HS":50,"SL":50,"IB":50,"CH":50,"BS":50},"coverag...
GT-013	Domain classification boundary 75	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":50,"SR":50,"HS":50,"SL":75,"IB":50,"CH":50,"BS":50},"coverag...
GT-014	Reverse-scored Q26 raw Always becomes zero burden	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":50,"SR":50,"HS":44,"SL":50,"IB":50,"CH":50,"BS":50},"coverag...
GT-015	Reverse-scored Q28 raw Never becomes maximum burden	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":50,"SR":50,"HS":56,"SL":50,"IB":50,"CH":50,"BS":50},"coverag...
GT-016	Exactly 50% coverage for four-item SL domain remains calculable	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":50,"SR":50,"HS":50,"SL":50,"IB":50,"CH":50,"BS":50},"coverag...
GT-017	Below 50% coverage makes SL null	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":50,"SR":50,"HS":50,"SL":null,"IB":50,"CH":50,"BS":50},"cover...
GT-018	N/A excluded from numerator and denominator	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":50,"SR":50,"HS":50,"SL":50,"IB":50,"CH":50,"BS":50},"coverag...
GT-019	Five available domains permit Biological State	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":null,"SR":null,"HS":50,"SL":50,"IB":50,"CH":50,"BS":50},"cov...
GT-020	Four available domains make Biological State null	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":null,"SR":null,"HS":null,"SL":50,"IB":50,"CH":50,"BS":50},"c...
GT-021	Co-primary driver tie within three points	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":0,"SR":100,"HS":0,"SL":100,"IB":0,"CH":0,"BS":0},"coverage":...
GT-022	No driver when every domain is below 25	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":0,"SR":0,"HS":0,"SL":0,"IB":0,"CH":0,"BS":0},"coverage":{ "MR...
GT-023	Low answer confidence reduces ROOTS Confidence	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":50,"SR":50,"HS":50,"SL":50,"IB":50,"CH":50,"BS":50},"coverag...
GT-024	High answer confidence and consistent responses	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":25,"SR":25,"HS":25,"SL":25,"IB":25,"CH":25,"BS":25},"coverag...
GT-025	Young age, no conditions, no medicines, five protective factors	{ "age":25,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":50,"SR":50,"HS":50,"SL":50,"IB":50,"CH":50,"BS":50},"coverag...
GT-026	Older age, multiple conditions and medicines, no protective factors	{ "age":80,"diseaseCount":3,"medicationCount":2,"P1":false,"P2":false,"P3":fal...	{ "domains":{ "MR":50,"SR":50,"HS":50,"SL":50,"IB":50,"CH":50,"BS":50},"coverag...
GT-027	Activity minimum burden mapping	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":29,"SR":50,"HS":50,"SL":50,"IB":50,"CH":50,"BS":50},"coverag...
GT-028	Activity maximum burden mapping	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":71,"SR":50,"HS":50,"SL":50,"IB":50,"CH":50,"BS":50},"coverag...
GT-029	One missing item in every domain	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":50,"SR":50,"HS":50,"SL":50,"IB":50,"CH":50,"BS":50},"coverag...
GT-030	Mixed high-burden profile with complete data	{ "age":40,"diseaseCount":0,"medicationCount":0,"P1":true,"P2":true,"P3":true,...	{ "domains":{ "MR":75,"SR":25,"HS":100,"SL":100,"IB":75,"CH":50,"BS":100},"cove...

QA_Checks

Check	Formula	Expected	Result
Domain count	7	7	PASS
Mapping count	40	40	PASS
Reverse-scored count	2	2	PASS
Golden tests	30	30	PASS
MR items	7	7	PASS
HS items	8	8	PASS
Classification rows	12	12	PASS
Driver outputs conform to DRV-001/002/003	30	30	PASS
No driver duplicates / ineligible fallback	30	30	PASS
BIO_STATE identifier unambiguous	1	1	PASS
Rounding rule unified	1	1	PASS